

# **CI12338tl-b – USB2.0 PHY**

LS-FS-HS 3.3/1.2/1.2 UTMI+ Level 3

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## Version History

| Version | Date       | Prepared | Description     |
|---------|------------|----------|-----------------|
| 1v0     | 2006-10-16 | TPrata   | Product Release |

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## 1 Features

- Complete PHY for USB2.0
- UTMI+ Level 3 specification compliant
- Supports 480Mbps “High Speed” (HS), 12Mbps “Full Speed” (FS) and 1.5Mbps “Low Speed” (LS) serial data transmission
- USB2.0 PHY for integration in HOST, DEVICE and Dual Role Device applications
- 8-bit and 16-bit parallel interface
- Built In Self Test (BIST)
- Integrated Self-calibrated termination resistors
- 3.3V and 1.2V  $\pm 10\%$  Analog supply operation
- 1.2V  $\pm 10\%$  Digital supply operation
- Core Area: 0.89 mm<sup>2</sup>
- Consumption in HS TX Mode: 45mA

### APPLICATIONS

- Digital video camera
- Mass storage device

## 2 General Description

USB2.0 PHY is a complete transceiver macro-cell that implements the USB2.0 physical layer.

It includes the following blocks:

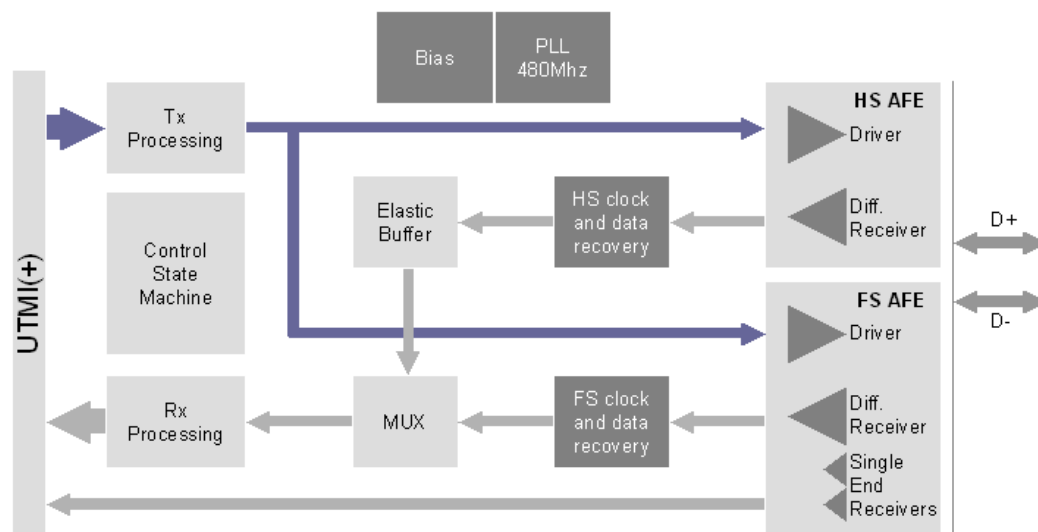
- Analog driver and receivers
- PLL to generate the 480MHz clock
- Bias current generator and voltage band-gap
- Clock and Data recovery
- Sync detector
- NRZI encoder/decoder
- Serialize/ De-serialize
- Control state machine

The transmitter uses Raw data in byte format that is converted into serial then a SYNC pattern is attached and is given to the analog front-end that drives the required pattern on the USB data lines. Meanwhile this data was bit-stuffed then NRZI encoded and an EOP pattern attached to it.

The receiver hunts for a SYNC pattern, performs the NRZI decoding, bit de-stuffing and serial to parallel conversion. The clock and data are recovered from the received serial stream at FS, while at HS the receiver operates on the recovered clock. The received data is given, to the upper layer, in byte format. There are additional functions like looking for bit-stuff and alignment errors and disconnect detection.

TECHNOLOGY: TSMC 0.13um Generic Digital CMOS process; 1Poly; 4 up to 6 Metals layers

## 3 Functional Diagram



## 4 Specifications

### OVERALL

| Parameter                      | Conditions | MIN | TYP | MAX  | Unit |
|--------------------------------|------------|-----|-----|------|------|
| Operating Junction Temperature |            | -40 | +25 | +125 | °C   |
| Analog Supply Voltage          |            | 3.0 | 3.3 | 3.6  | V    |
| Analog Supply Voltage          |            | 1.1 | 1.2 | 1.3  | V    |
| Digital Supply Voltage         |            | 1.1 | 1.2 | 1.3  | V    |

### Current Consumption

| Parameter      | Conditions   | MIN | TYP | MAX | Unit |
|----------------|--------------|-----|-----|-----|------|
| Supply Current | Idle         |     | 15  | 25  | mA   |
| Supply Current | Full Speed   |     | 30  | 40  | mA   |
| Supply Current | High Speed   |     | 45  | 55  | mA   |
| Supply Current | Suspend Mode |     |     | 50  | uA   |

### Timing Characteristics

| Parameter | Conditions | MIN | TYP | MAX | Unit |
|-----------|------------|-----|-----|-----|------|
| trise     | 1,5Mbps    | 75  |     | 300 | ns   |
|           | 12Mbps     | 4   |     | 20  |      |
|           | 480Mbps    | 0,5 |     |     |      |
| tfall     | 1,5Mbps    | 75  |     | 300 | ns   |
|           | 12Mbps     | 4   |     | 20  |      |
|           | 480Mbps    | 0,5 |     |     |      |
| Jitter    | 1,5Mbps    |     |     | 10  | ns   |
|           | 12Mbps     |     |     | 1   |      |
|           | 480Mbps    |     |     | 0,2 |      |

### Electrical Characteristics

| Parameter                                           | Conditions            | MIN          | TYP | MAX        | Unit |
|-----------------------------------------------------|-----------------------|--------------|-----|------------|------|
| Vcm DC<br>(DC level measured at receiver connector) | HS Mode<br>LS/FS Mode | -0,05<br>0,8 |     | 0,5<br>2,5 | V    |
| Crossover Voltages                                  | LS Mode<br>FS Mode    | 1,3<br>1,3   |     | 2<br>2     | V    |
| Power supply ripple noise<br>(Analog 3.3V)          | <160MHz               | -50          | 0   | 50         | mV   |
| Power supply ripple noise PLL<br>(Analog 1.2V)      | < 4MHz                | -10          | 0   | 10         | mV   |
|                                                     | > 4MHz                | -50          | 0   | 50         |      |
| Power supply ripple noise CORE<br>(Digital 1.2V)    | All conditions        | -50          | 0   | 50         | mV   |

### System Clocking (sysclock)

| Parameter       | Conditions | MIN  | TYP | MAX | Unit |
|-----------------|------------|------|-----|-----|------|
| Clock deviation |            | -200 |     | 200 | ppm  |
| Rise/fall time  |            |      |     | 200 | ps   |
| Jitter          |            |      |     | 50  | ps   |
| Duty-Cycle      |            | 40   |     | 60  | %    |

## 5 Pin Description

### Power

| Pin Name    | I/O | Type   | Function                |
|-------------|-----|--------|-------------------------|
| <i>vdda</i> | P   | Power  | Analog core 3.3V supply |
| <i>vssa</i> | G   | Ground | Analog core ground pin  |
| <i>vddp</i> | P   | Power  | PLL 1.2V supply         |
| <i>vssp</i> | G   | Ground | PLL ground pin          |
| <i>vddd</i> | P   | Power  | Digital 1.2V supply     |
| <i>vssd</i> | G   | Ground | Digital Ground pin      |

### USB Signals (Cable)

| Pin Name  | I/O | Type   | Function                                                           |
|-----------|-----|--------|--------------------------------------------------------------------|
| <i>dp</i> | I/O | Analog | Positive output channel that is connected to the serial USB cable. |
| <i>dn</i> | I/O | Analog | Negative output channel that is connected to the serial USB cable. |

### System Clocks

| Pin Name           | I/O | Type    | Function                                                                                                                                                                                                                                                                                           |
|--------------------|-----|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>sysclock</i>    | I   | Digital | This input is the reference clock source for the control circuit and is the basis from which the transmitting interface speed is established. <i>SYSCLOCK</i> is used by the PLL to generate the 480MHz clock and the <i>SIECLOCK</i> signal. Its nominal frequency depends on <i>setin12</i> pin. |
| <i>sieclock</i>    | O   | Digital | This output is used for clocking receive and transmit parallel data:<br>. 60 MHz HS/FS, with 8-bit interface<br>. 30 MHz HS/FS, with 16-bit interface                                                                                                                                              |
| <i>databus16_8</i> | I   | Digital | Selects between 8 and 16 bit data transfers.<br>"1"- 16-bit data path operation enabled. <i>SIECLOCK</i> = 30 MHz.<br>"0"- 8-bit data path operation enabled. <i>SIECLOCK</i> = 60 MHz.                                                                                                            |
| <i>setin12</i>     | I   | Digital | Selects sysclock reference value<br>"1"- 12MHz<br>"0"- 24MHz                                                                                                                                                                                                                                       |

### Testability Signals

| Pin Name                   | I/O | Type    | Function                                                                   |
|----------------------------|-----|---------|----------------------------------------------------------------------------|
| <i>vload</i>               | I   | Digital | Active low.<br>Assertion of this signal loads the Vendor Control register. |
| <i>vcontrol&lt;3,0&gt;</i> | I   | Digital | Vendor Control - Chipidea defined 4-bit parallel input.                    |
| <i>vstatus&lt;7,0&gt;</i>  | O   | Digital | Vendor Status - Chipidea defined 8-bit parallel output.                    |
| <i>onbist</i>              | I   | Digital | Active High<br>Assertion of this signal start bist mode                    |
| <i>gpanaio</i>             | O   | Analog  | Observability of Internal Analog Signals                                   |
| <i>vbg</i>                 | O   | Analog  | Bandgap reference voltage                                                  |
| <i>ibiascp</i>             | O   | Analog  | Should be left unconnected                                                 |

### Scan Test Signals

| Pin Name        | I/O | Type    | Function                                |
|-----------------|-----|---------|-----------------------------------------|
| <i>scanmode</i> | I   | Digital | Active High.<br>Enables scan mode test. |
| <i>scanen</i>   | I   | Digital | Active High.<br>Scan enable.            |
| <i>scanclk</i>  | I   | Digital | Scan clock.                             |
| <i>scanina</i>  | I   | Digital | Scan input for scan-chain a.            |
| <i>scaninb</i>  | I   | Digital | Scan input scan-chain b.                |
| <i>scanouta</i> | O   | Digital | Scan output scan-chain a.               |
| <i>scanoutb</i> | O   | Digital | Scan output scan-chain b.               |

### External Components

| Pin Name       | I/O | Type   | Function                                           |
|----------------|-----|--------|----------------------------------------------------|
| <i>rrefext</i> | I/O | Analog | External resistor connection for current reference |

### Control/Status Signal

| Pin Name                    | I/O | Type    | Function                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|-----|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>reset</i>                | I   | Digital | Active High. Asynchronous.<br>Reset all state machines in the UTM to its default condition.                                                                                                                                                                                                                                                                                                                     |
| <i>suspend</i>              | I   | Digital | Active Low. Asynchronous. (Macrocell circuitry drawing suspend current)<br>Places the Macrocell in a mode that draws minimal power from supplies. Shuts down all blocks not necessary for Suspend/Resume operation. While suspended, <i>TERMSEL</i> must always be in FS mode to ensure that the 1.5K pull-up on <i>DP</i> remains powered.                                                                     |
| <i>xcvrse/ &lt;1:0&gt;</i>  | I   | Digital | This signal selects between the LS, FS and HS transceivers:<br>"00"- HS transceiver enabled<br>"01"- FS transceiver enabled.<br>"10"- LS transceiver enable<br>"11"- Send a LS packet on a FS bus or receive a LS packet                                                                                                                                                                                        |
| <i>termse/</i>              | I   | Digital | This signal selects between the FS and HS terminations:<br>"0"- HS termination enabled<br>"1"- FS termination enabled.                                                                                                                                                                                                                                                                                          |
| <i>opmode&lt;1:0&gt;</i>    | I   | Digital | These signals select between various operational modes<br>"00" - Normal Operation<br>"01" - Non-Driving<br>"10" - Disable Bit Stuffing and NRZI encoding<br>"11" - Normal operation without automatic generation of SYNC and EOP. Only valid when <i>XCVRSEL</i> = "00".                                                                                                                                        |
| <i>linestate&lt;1:0&gt;</i> | O   | Digital | These signals reflect the current state of the single ended receivers. They are combinatorial until a "usable" <i>SIECLOCK</i> is available then they are synchronized to <i>SIECLOCK</i> . They directly reflect the current state of the <i>DP</i> (LineState[0]) and <i>DN</i> ( <i>LINESTATE</i> [1]) signals:<br><b>DN DP Description</b><br>0 0 - SE0<br>0 1 - 'J' State<br>1 0 - 'K' State<br>1 1 - SE1. |
| <i>dppulldown</i>           | I   | Digital | Control DP line pull down resistor. Active High.                                                                                                                                                                                                                                                                                                                                                                |
| <i>dnpulldown</i>           | I   | Digital | Control DN line pull down resistor. Active High.                                                                                                                                                                                                                                                                                                                                                                |
| <i>hostdisconnect</i>       | O   | Digital | Indicates peripheral disconnect detection. Only valid if <i>DPPULLDOWN</i> and <i>DNPULLDOWN</i> are asserted.                                                                                                                                                                                                                                                                                                  |
| <i>txbitstufferable</i>     | I   | Digital | Active High. During <i>OPMODE</i> =11 indicates data on the <i>DATAIN</i> (7:0) lines needs to be bitstuffed.                                                                                                                                                                                                                                                                                                   |
| <i>txbitstufferableH</i>    | I   | Digital | Active High. During <i>OPMODE</i> =11 indicates data on the <i>DATAIN</i> (15:8) lines needs to be bitstuffed.                                                                                                                                                                                                                                                                                                  |
| <i>conf1</i>                | I   | Digital | Tie low                                                                                                                                                                                                                                                                                                                                                                                                         |
| <i>conf2</i>                | I   | Digital | Tie low                                                                                                                                                                                                                                                                                                                                                                                                         |

### Transmit Channel

| Pin Name                  | I/O | Type    | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------|-----|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>datain&lt;15:0&gt;</i> | I   | Digital | Data sent from the Serial Interface Engine (SIE) to the Phy layer for serialization and transmission.<br>The transmitter accepts 8 or 16 bit data words as inputs to the serializer on the <i>DATAIN</i> inputs. If <i>DATABUS16_8</i> =0 all transmit data is transferred over the lower 8 bits bus. Note that <i>DATAIN</i> bit 0 is transmitted first.                                                                                                                                                                                                                                                                                                                                                                        |
| <i>txvalid</i>            | I   | Digital | Active HIGH.<br>Indicates that the <i>DATAIN</i> bus is valid. The assertion of Transmit Valid initiates SYNC on the USB. The negation of Transmit Valid initiates EOP on the USB. In HS ( <i>XCVRSEL</i> =0,0) mode, the SYNC pattern must be asserted on the USB between 8 and 16 bit times after the Transmit State Machine detects the assertion of <i>TXVALID</i> . In FS ( <i>XCVRSEL</i> =01) or LS, the SYNC pattern must be asserted on the USB no less than 1 Clock and no more than 5 - 10 Clocks after the assertion of <i>TXVALID</i> is detected by the Transmit State Machine.                                                                                                                                    |
| <i>txvalidh</i>           | I   | Digital | Active HIGH.<br>When <i>DATABUS16_8</i> =1, this signal indicates that the <i>DATAIN</i> (8,15) bus contains valid transmit data. This signal is ignored when <i>DATABUS16_8</i> =0..                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <i>txready</i>            | O   | Digital | Active HIGH.<br>If <i>TXVALID</i> is asserted, the SIE must always have data available for clocking in to the TX Holding Register on the rising edge of <i>SIECLOCK</i> . If <i>TXVALID</i> =1 and <i>TXREADY</i> is asserted at the rising edge of <i>SIECLOCK</i> , the UTM will load the data on the <i>DATAIN</i> bus into the TX Holding Register on the next rising edge of <i>SIECLOCK</i> , at that time, SIE should immediately present the data for next transfer on the <i>DATAIN</i> bus. If <i>TXVALID</i> is asserted and <i>TXREADY</i> is negated, the SIE must hold the previously asserted data on the <i>DATAIN</i> bus. From the time <i>TXVALID</i> is negated, <i>TXREADY</i> is a don't care for the SIE. |

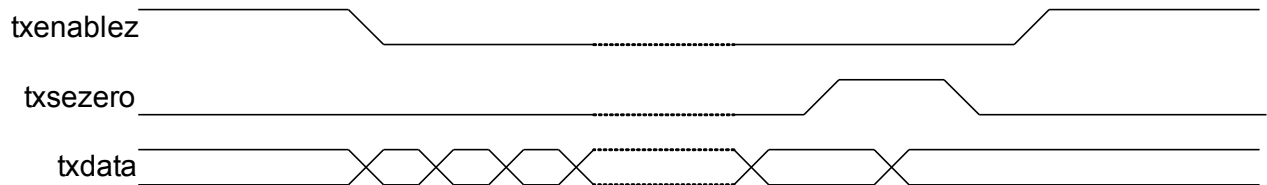
### Receive Channel

| Pin Name                   | I/O | Type    | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|-----|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>dataout&lt;15:0&gt;</i> | O   | Digital | All receive data is transferred over this bus. When <i>DATABUS16_8</i> =0 only the low byte of the <i>DATAOUT</i> bus is used.<br><i>DATAOUT</i> bit 0 is the earliest bit received at the RX inputs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <i>rxvalid</i>             | O   | Digital | Active HIGH.<br>Indicates that the <i>DATAOUT</i> bus has valid data. The Receive Data Holding Register is full and ready to be unloaded. The SIE is expected to latch the <i>DATAOUT</i> bus on the clock edge.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <i>rxvalidh</i>            | O   | Digital | Active HIGH.<br>When <i>DATABUS16_8</i> =1 this signals indicates that the <i>DATAOUT</i> (8-15) bus is presenting valid receive data. This signal is ignored when <i>DATABUS16_8</i> =0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <i>rxactive</i>            | O   | Digital | Active HIGH. Indicates that the receive state machine has detected SYNC and is active. <i>RXACTIVE</i> is negated after a Bit Stuff Error or an EOP is detected. In HS mode ( <i>XCVRSEL</i> =0), <i>RXACTIVE</i> must be negated no less than 3 and no more than 8 <i>SIECLOCK</i> after an Idle state is detected on the USB. And <i>RXACTIVE</i> must be negated for at least 1 <i>SIECLOCK</i> between consecutive received packets. In FS ( <i>XCVRSEL</i> =1), FS Only, or LS Only modes, <i>RXACTIVE</i> must be negated no more than 2 <i>SIECLOCK</i> after a FS Idle state is detected on the USB. And <i>RXACTIVE</i> must be negated for at least 4 <i>SIECLOCK</i> between consecutive received packets. |
| <i>rxerror</i>             | O   | Digital | Active HIGH (error detected).<br>This output is clocked with the same timing as the <i>DATAOUT</i> lines and can occur at anytime during a transfer. If asserted, it will force the negation of <i>RXVALID</i> on the next rising edge of <i>SIECLOCK</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

#### FS/FS Serial Mode

| Pin Name              | I/O | Type    | Function                                                                                                                       |
|-----------------------|-----|---------|--------------------------------------------------------------------------------------------------------------------------------|
| <i>txsezero</i>       | I   | Digital | Force Single-Ended Zero on the bus                                                                                             |
| <i>fslsserialmode</i> | I   | Digital | "0" - FS and LS packets are sent using the parallel interface.<br>"1" - FS and LS packets are sent using the serial interface. |
| <i>txenablez</i>      | I   | Digital | Active low input enable signal.                                                                                                |
| <i>txdata</i>         | I   | Digital | Serial data to be driven on <i>DP</i> and <i>DN</i> .                                                                          |
| <i>rxdp</i>           | O   | Digital | Single-ended receive data, positive terminal.<br>The data is only valid if <i>FSLSSESRIALMODE</i> is set to "1"                |
| <i>rxdm</i>           | O   | Digital | Single-ended receive data, negative terminal.<br>The data is only valid if <i>FSLSSESRIALMODE</i> is set to "1"                |
| <i>rxrcv</i>          | O   | Digital | Receive data. The data is only valid if <i>FSLSSESRIALMODE</i> is set to "1"                                                   |

*txdata* and *txsezero* signals shouldn't be asserted at the same time; it is recommended that two SE0 bit times are implemented in the EOP and, if it is necessary to change the *txdata* value from K to J for driving the last EOP bit, this is done in the middle of the two SE0 bits; please refer to the figure below for more details.



## 6 Operating Modes

### Operation Modes

| Mode                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Normal Operation</b>                                | Activated by the <i>OPMODE</i> = "0,0".<br>Allows the transceiver to operate with normal USB data decoding and encoding                                                                                                                                                                                                                                                                               |
| <b>Non Driving</b>                                     | Activated by the <i>OPMODE</i> = "0,1".<br>Allows the transceiver logic to support a soft disconnect feature which tri-states both the HS and FS transmitters, and removes any termination from the USB making it appear to an upstream port that the device has been disconnected from the bus.                                                                                                      |
| <b>Disable Bit Stuffing and NrZI encoding</b>          | Activated by the <i>OPMODE</i> = "1,0".<br>Disables Bit Stuff and NRZI encoding logic so 1's loaded from the <i>DATAIN</i> bus becomes 'J's on the <i>DP/DN</i> lines and 0's become 'K's.<br>Note that this mode affects the automatic SYNC Pattern and EOP generation by <i>TXVALID</i> . It is disabled so that Chirps can be generated on the USB.<br>The operation of the receiver is undefined. |
| <b>Normal operation without automatic SYNC and EOP</b> | Activated by the <i>OPMODE</i> = "1,1".<br>NRZI encoding is always enabled. Bit stuffing depends on the value of <i>TxBitstuffEnable</i> and <i>TxBitstuffEnableH</i> . This is only valid when <i>XcvtSelect</i> is set to 00b. If <i>OpMode</i> is set to 11b together with <i>XcvtSelect</i> not equal to 00b, the behavior of the transceiver is undefined.                                       |

### Speed Selection

| Mode      | Description                                                                                                                                                                                                                                                 |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HS</b> | Activated by the <i>XCVRSEL</i> = 00.<br>The HS Detection Handshake protocol requires independent control of transceivers and terminations, where the device enables FS terminations but is required to drive and listen to the bus with the HS transceiver |
| <b>FS</b> | Activated by the <i>XCVRSEL</i> = 01.<br>And, enabling the respective termination.                                                                                                                                                                          |
| <b>LS</b> | Activated by the <i>XCVRSEL</i> = 10.<br>And, enabling the respective termination                                                                                                                                                                           |

### Power States

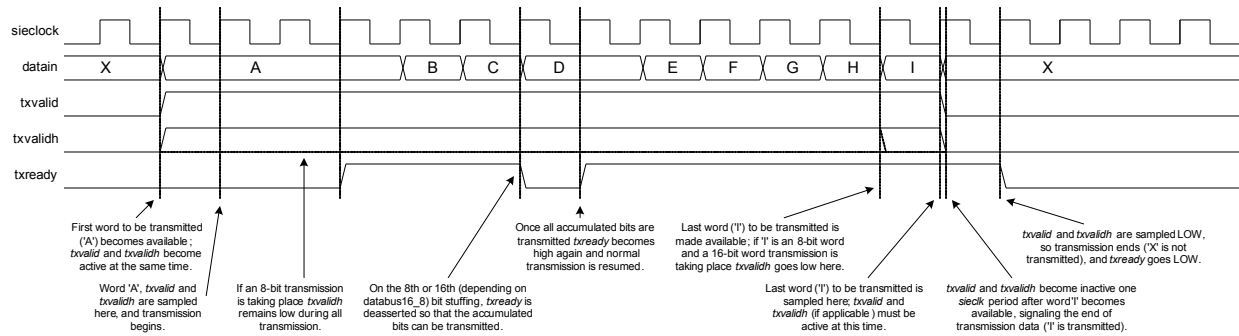
| Mode                    | Description                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------|
| <b>Normal Operation</b> | Default state after <i>RESET</i> .<br>The PHY logic and main PLL are both on and active.      |
| <b>Suspend</b>          | Activated by <i>SUSPEND</i> pin.<br>The PHY logic is powered but is in a reduced power state. |

## 7 Signals Timing

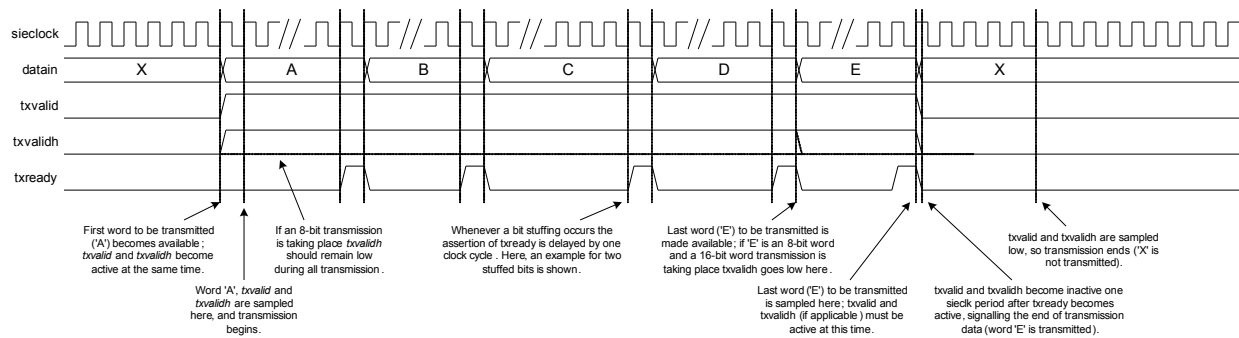
### Low/Full-Speed and High-Speed Receive and Transmit Timings

Shown below are the Low/Full-Speed and High-Speed receive and transmit timing diagrams.

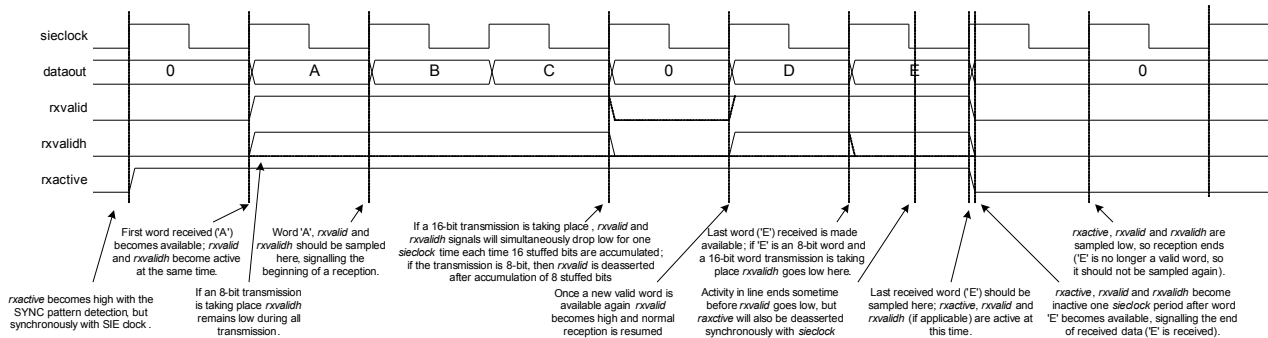
#### High-Speed Tx



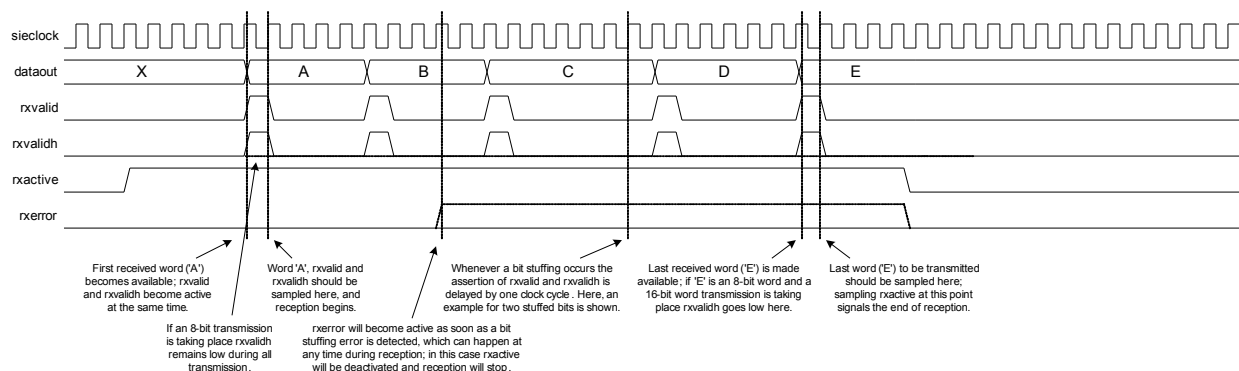
#### Full/Low-Speed Tx



#### High-Speed Rx

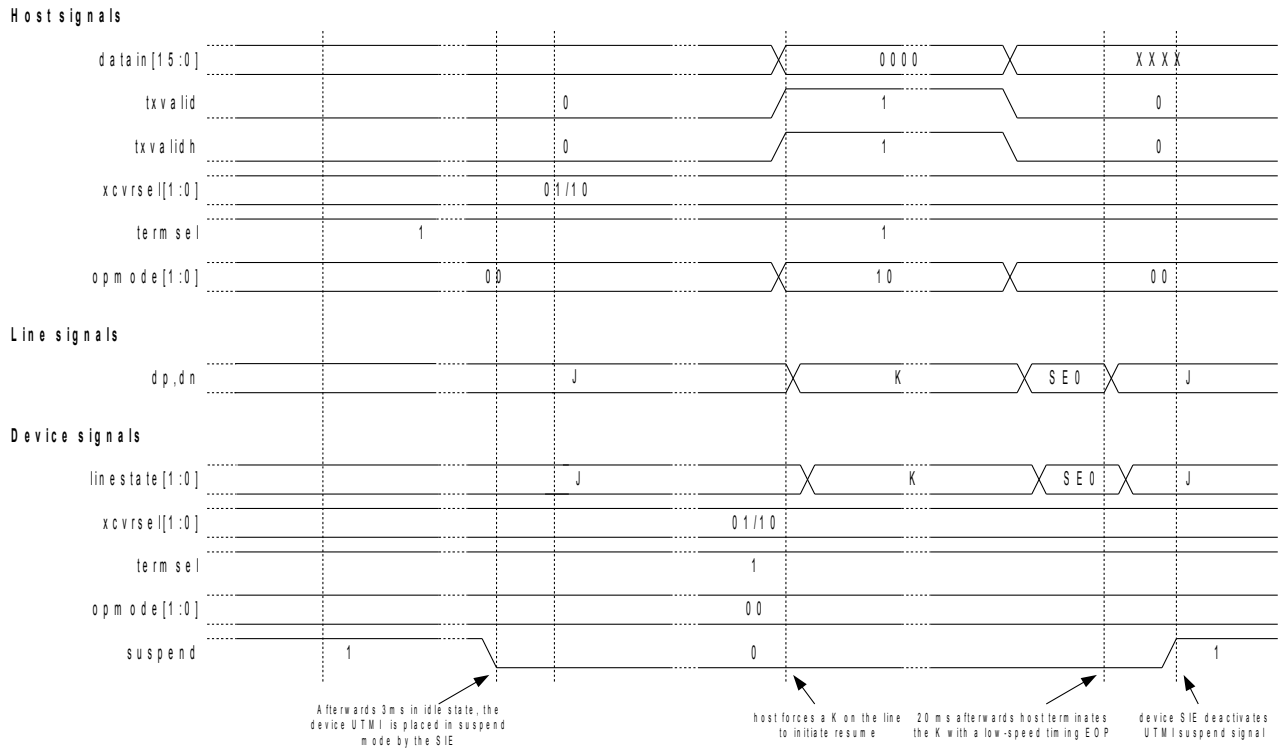


#### Full/Low-Speed Rx

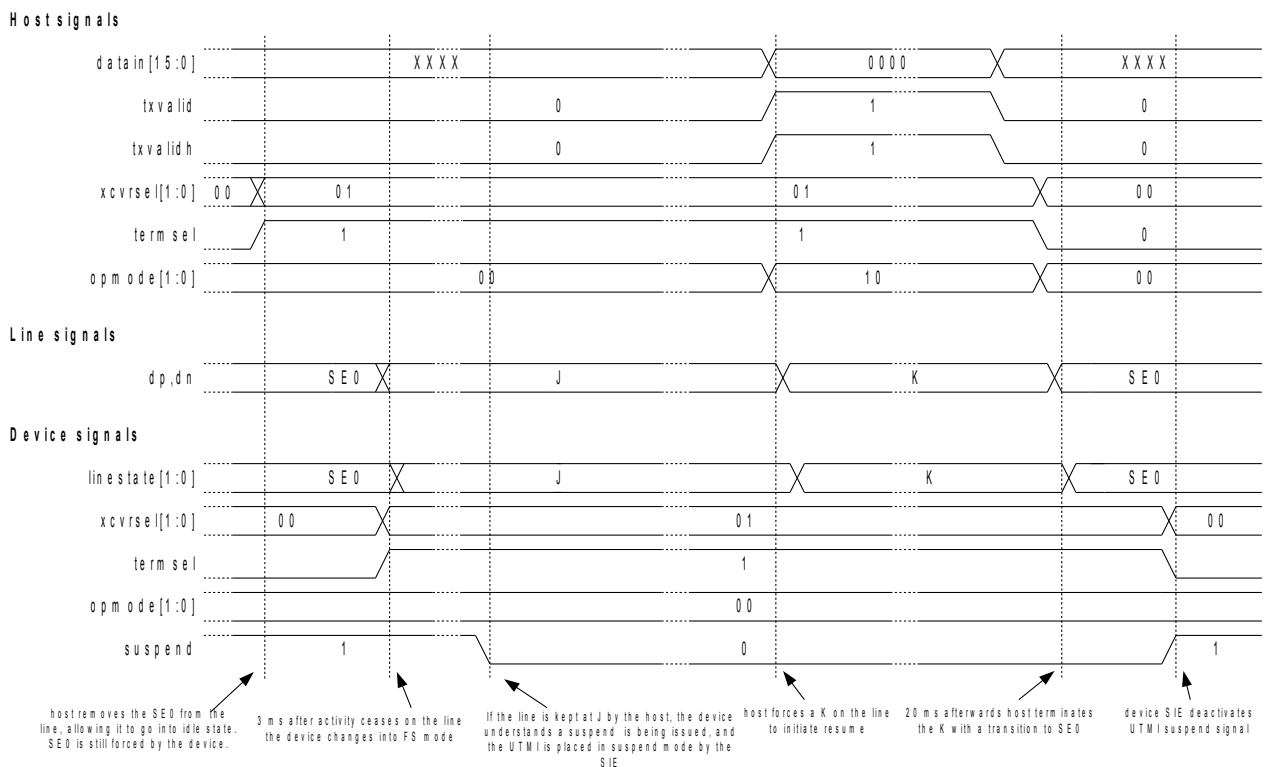


Shown below are the timing diagrams for suspend and resume signaling, from host and device side.

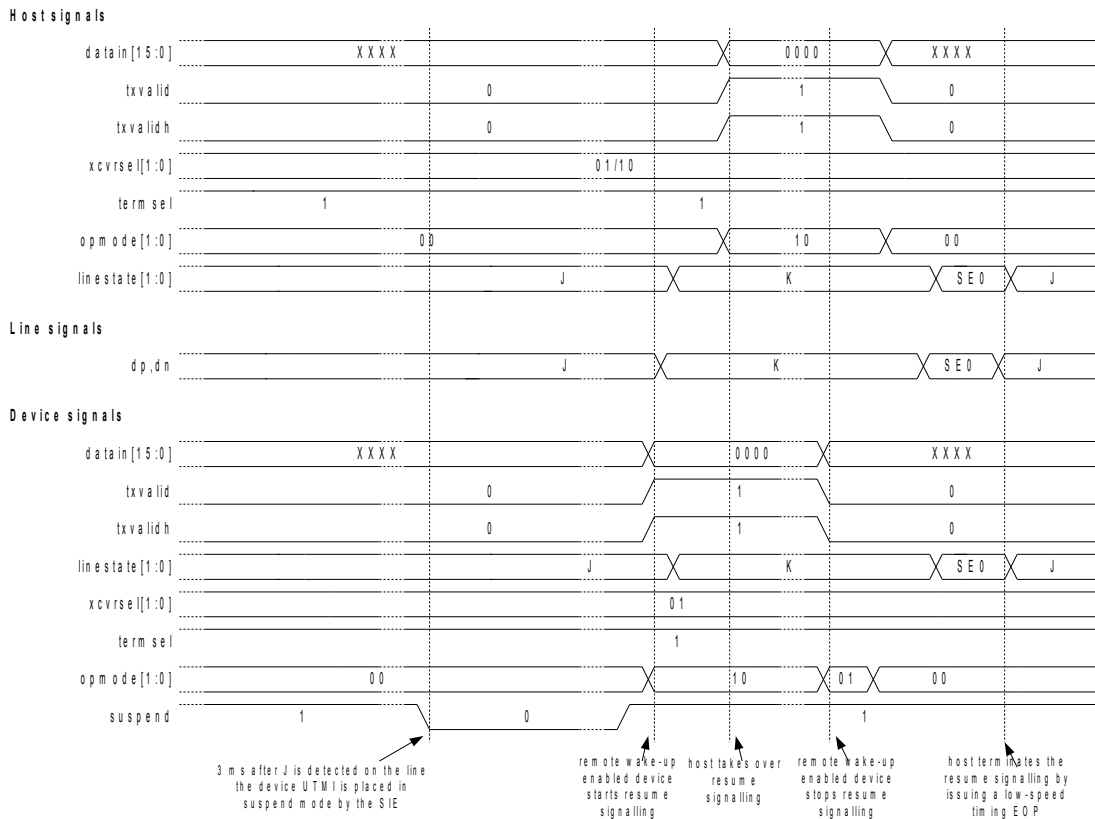
### Low/Full speed suspend and resume timing.



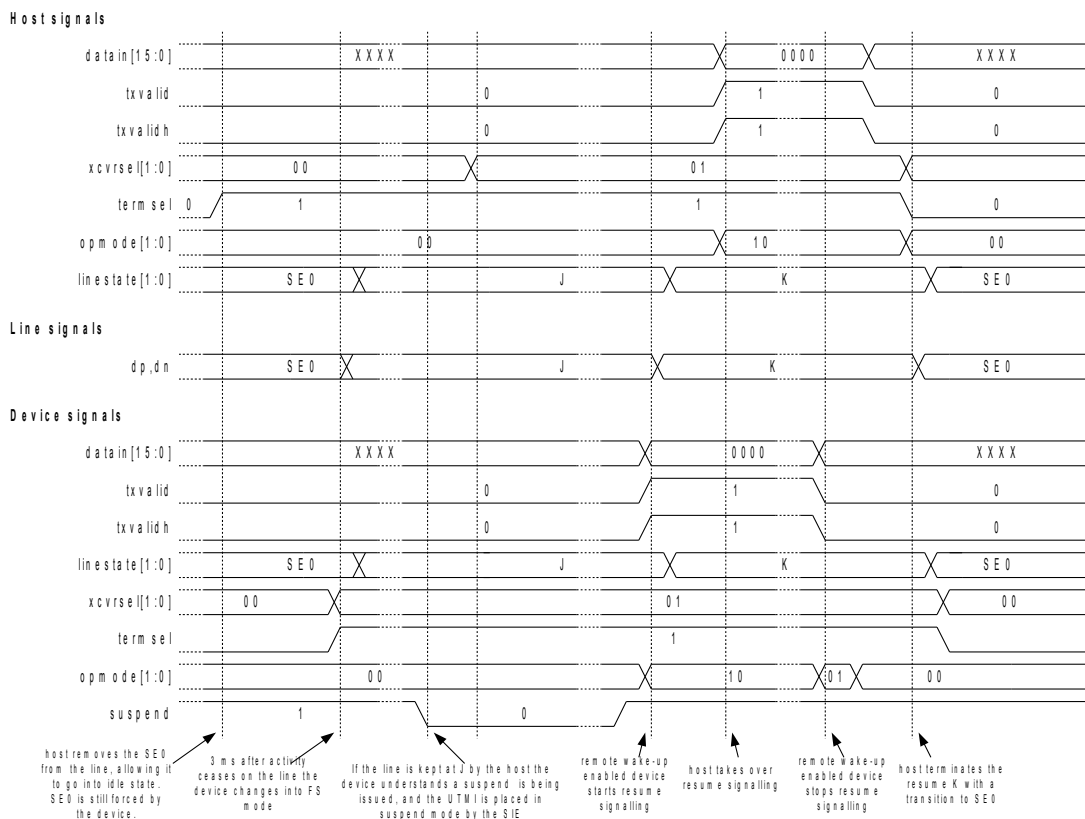
### High-Speed suspend and resume signaling.



## Low/Full-Speed suspend and resume signaling with remote wake-up.

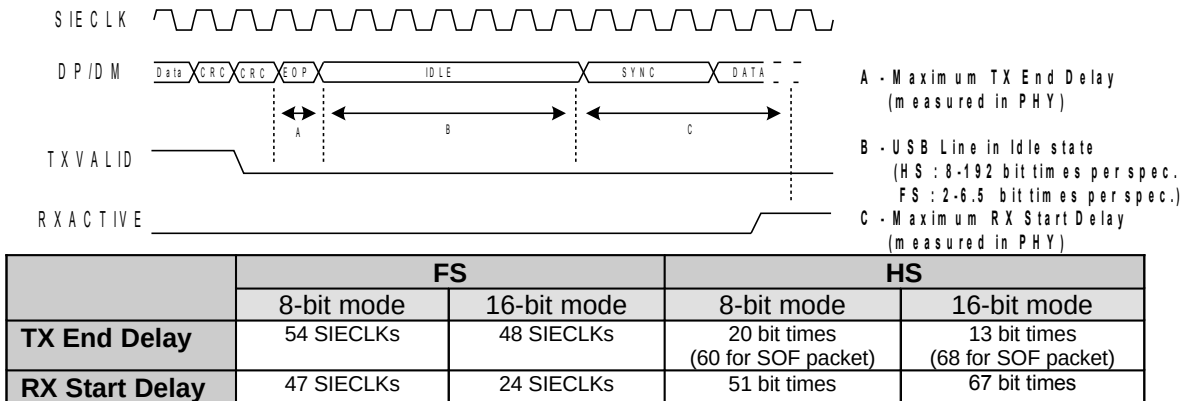


## High-Speed suspend and resume signaling with remote wake-up.



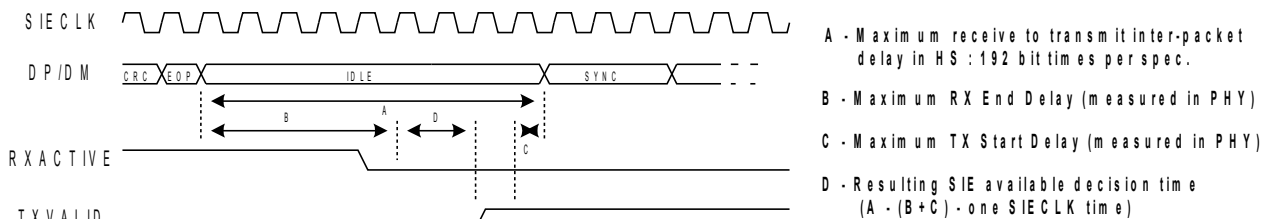
### Transmit to Receive Inter-packet Delays

Inter-packet delay for a Transmit followed by a Receive. In this case the downstream port determines the inter-packet delay.



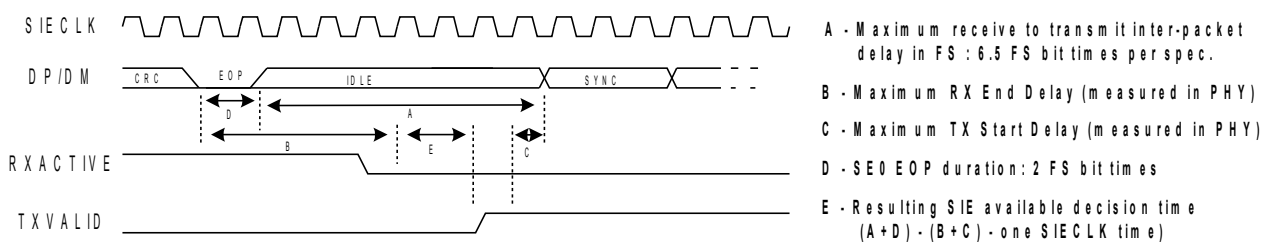
### Receive to Transmit Inter-packet Delays

Inter-packet delay for a receive followed by a transmit – **HS mode**



Receive to Transmit Inter-packet Delays as measured in HS mode

Inter-packet delay for a receive followed by a transmit – **FS mode**



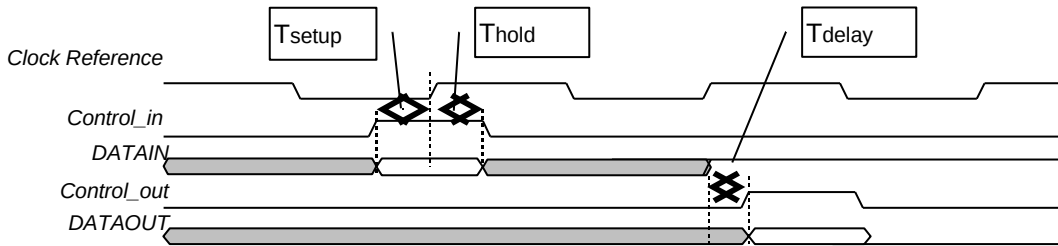
Receive to Transmit Inter-packet Delays as measured in FS mode

|                       | FS         |             | HS           |              |
|-----------------------|------------|-------------|--------------|--------------|
|                       | 8-bit mode | 16-bit mode | 8-bit mode   | 16-bit mode  |
| <b>TX Start Delay</b> | 3 SIECLKs  | 2 SIECLKs   | 12 bit times | 12 bit times |
| <b>RX End Delay</b>   | 17 SIECLKs | 9 SIECLKs   | 47 bit times | 85 bit times |

### Receive to Receive Inter-packet Delays

|                           | FS          |             | HS           |              |
|---------------------------|-------------|-------------|--------------|--------------|
|                           | 8-bit mode  | 16-bit mode | 8-bit mode   | 16-bit mode  |
| <b>Inter-Packet Delay</b> | 2 bit times | 2 bit times | 32 bit times | 32 bit times |

## Timing Constraints



## Timing Constraints Conditions

| clock domain | Tsetup         |                  | Thold          |                  | unit |
|--------------|----------------|------------------|----------------|------------------|------|
|              | pin transition | clock transition | pin transition | clock transition |      |
| sieclock     | 5.0            | 0                | 0              | 5.0              | ns   |
| sysclock     | 5.0            | 0                | 0              | 5.0              | ns   |
| scanclk      | 5.0            | 0                | 0              | 5.0              | ns   |

## Timing Constraints of Input Signal

| Control In<br>Data In | Clock<br>Reference | Tsetup |     |     | Thold |     |     | Unit |
|-----------------------|--------------------|--------|-----|-----|-------|-----|-----|------|
|                       |                    | min    | typ | max | min   | typ | max |      |
| xcvrsel[1:0]          | sieclock           | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| datain[15:0]          | sieclock           | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| termssel              | sieclock           | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| databus16_8           | asynchronous       | -      | -   | -   | -     | -   | -   | ns   |
| txvalidh              | sieclock           | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| txvalid               | sieclock           | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| vload                 | sysclock           | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| vcontrol[3:0]         | sysclock           | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| scanmode              | asynchronous       | -      | -   | -   | -     | -   | -   | ns   |
| scanen                | scanclk            | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| scanin                | scanclk            | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| reset                 | asynchronous       | -      | -   | -   | -     | -   | -   | ns   |
| suspend               | asynchronous       | -      | -   | -   | -     | -   | -   | ns   |
| opmode[1:0]           | sieclock           | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| dppulldown            | sieclock           | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| dnpulldown            | sieclock           | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| txbitstufferable      | sieclock           | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| txbitstufferableH     | sieclock           | 5.0    | -   | -   | 0.0   | -   | -   | ns   |
| txsezero              | asynchronous       | -      | -   | -   | -     | -   | -   | ns   |
| fslsserialmode        | asynchronous       | -      | -   | -   | -     | -   | -   | ns   |
| txenablez             | asynchronous       | -      | -   | -   | -     | -   | -   | ns   |
| txdata                | sieclock           | -      | -   | -   | -     | -   | -   | ns   |

## Timing Constraints of Output Signal

| Control Out<br>Data Out | Clock<br>Reference | Conditions           |                              | Tdelay |     |     | Unit |
|-------------------------|--------------------|----------------------|------------------------------|--------|-----|-----|------|
|                         |                    | input net transition | total output net capacitance | min    | typ | max |      |
| rxerror                 | sieclock           | 5.0000               | 1                            | -      | -   | 5.0 | ns   |
| rxactive                | sieclock           | 5.0000               | 1                            | -      | -   | 5.0 | ns   |
| rxvalidh                | sieclock           | 5.0000               | 1                            | -      | -   | 5.0 | ns   |
| rxvalid                 | sieclock           | 5.0000               | 1                            | -      | -   | 5.0 | ns   |
| txready                 | sieclock           | 5.0000               | 1                            | -      | -   | 5.0 | ns   |
| linestate[1:0]          | sieclock           | 5.0000               | 1                            | -      | -   | 5.0 | ns   |
| dataout[15:0]           | sieclock           | 5.0000               | 1                            | -      | -   | 5.0 | ns   |
| hostdisconnect          | sieclock           | 5.0000               | 1                            | -      | -   | 5.0 | ns   |
| rxdp                    | sieclock           | 5.0000               | -                            | -      | -   | -   | ns   |
| rxdm                    | sieclock           | 5.0000               | -                            | -      | -   | -   | ns   |
| rxrcv                   | sieclock           | 5.0000               | -                            | -      | -   | -   | ns   |

## 8 Test Modes

The USB Specification defines the following test modes:

To force an SE0 State on the bus the UTM is placed in *opmode* 0 (Normal Operation) and no data is transmitted. This results in an HS Idle mode on the bus, which is SE0.

To force a 'J' State on the bus the UTM is placed in *opmode* 2 (Disable Bit Stuffing and NRZI encoding) and all 1's are loaded into the Data In Bus.

To force a 'K' State on the bus the UTM is placed in *opmode* 2 (Disable Bit Stuffing and NRZI encoding) and all 0's are loaded into the Data In Bus.

To generate a Test Packet on the bus the UTM is placed *opmode* 0 (Normal Operation) and all the Test Packet data is successively loaded into the Data In Bus.

| USB2.0 Test Mode | OPMODE  | XCVRSEL TERMSEL | Data transmitted | Description                        |
|------------------|---------|-----------------|------------------|------------------------------------|
| SE0_NAK          | Normal  | HS              | No transmit      | HS Idle mode on the bus.           |
| J                | Disable | HS              | All "1"s         | Force a 'J' State on the bus.      |
| K                | Disable | HS              | All "0"s         | Force a 'K' State on the bus.      |
| Test Packet      | Normal  | HS              | Test Packet Data | Generate a Test Packet on the bus. |

### Testability of each block

All test modes are programmed by setting VLoad to 0 while in suspend mode for one cycle of the external crystal clock. The test mode **code** is read from VControl at this point. As soon as VLoad is set back to 1, VControl becomes usable for "tapping" internal core signals according to the programmed test mode.

| Block        | Code | Description                                                  |
|--------------|------|--------------------------------------------------------------|
| No one       | 0000 | No test.                                                     |
| Loopback     | 0001 | Loopback test mode.                                          |
| Rx chain     | 0010 | Reserved for Chipidea usage.                                 |
| Tx chain     | 0011 | Reserved for Chipidea usage.                                 |
| HSCDR        | 0100 | Reserved for Chipidea usage.                                 |
| RESCAL       | 0101 | Resistor calibration test mode.                              |
| HS Receiver  | 0110 | High Speed Receiver Front-end test mode.                     |
| LFS Receiver | 0111 | Low-Full Speed Receiver Front-end test mode.                 |
| HS Driver    | 1000 | High Speed Driver test mode.                                 |
| FS Driver    | 1001 | Full Speed Driver Front-end test mode.                       |
| LS Driver    | 1010 | Low Speed Driver Front-end test mode.                        |
| -            | 1011 | Reserved for Chipidea usage.                                 |
| PLL          | 1100 | PLL testmode.                                                |
| -            | 1101 | Reserved for future usage.                                   |
| LOCKCTRL     | 1110 | External resistor calibration and PLL lock control test mode |
| -            | 1111 | Reserved for future usage.                                   |

## Production Test

Through the ONBIST input signal, a series of BIST sequences designed mainly for production testing can be selected:

The implemented BIST sequences are the following:

| VCONTROL  | BIST Sequence                   |
|-----------|---------------------------------|
| 0000      | No test                         |
| 0001      | Integral loopback BIST, HS mode |
| 0010      | Integral loopback BIST, FS mode |
| 0011-1111 | Reserved for future             |

The integral loopback tests use an augmented version of the USB Standard Test Packet, to include a Start-Of-Frame (SOF) Packet ID. This allows self test of the HS Disconnect Envelope Detector. Without using any external load, the voltage on DP/DN in High Speed will be around 800mV, enough to trigger the disconnect detector at the end of the packet.